Lesson 13: Martingale Convergence Theory

AI/ML Mathematics

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1 Motivation and Overview

The convergence of stochastic approximation algorithms—TD learning, Q-learning, SARSA—ultimately rests on the convergence of certain non-negative supermartingales. In Lesson 8 we defined martingales and martingale difference sequences; now we develop the convergence theory that tells us *when* these objects converge almost surely.

The central result is Doob's forward convergence theorem: a submartingale whose positive part has uniformly bounded expectation converges almost surely. The proof proceeds through a beautiful counting argument—the *upcrossing inequality* —which bounds how often the process can oscillate between two levels. We then prove the Robbins–Siegmund theorem, which generalizes supermartingale convergence to "almost supermartingales" with summable perturbations. This is the single most important convergence tool in the stochastic approximation literature.

Why This Matters for RL

- **Convergence of value estimates:** In the convergence proof of TD(0), the squared error $\|V_t - V^\pi\|^2$ is shown to be (approximately) a non-negative supermartingale. Doob's convergence theorem guarantees that bounded non-negative supermartingales converge almost surely—so the estimation error converges.
- **Robbins–Siegmund as the master tool:** The Robbins–Siegmund theorem directly handles the perturbations (step-size noise, approximation error) that prevent $\|V_t - V^\pi\|^2$ from being an exact supermartingale. Nearly every convergence proof for stochastic approximation in RL invokes this theorem.
- **Convex functions of martingales:** The fact that $\{M_n^2\}$ is a submartingale when $\{M_n\}$ is a martingale connects to variance analysis: the expected squared error of a martingale estimator can only grow, motivating variance reduction techniques (baselines in REINFORCE, control variates).

Prerequisites. Lesson 8 (filtrations, martingales, MDS, submartingales, supermartingales), Lesson 9 (Jensen's inequality, Markov's inequality), Lesson 11 (monotone convergence theorem, Fatou's lemma).

2 References

The following sources complement this lesson:

- Williams [1991], Chapters 11–12: a clean and accessible treatment of the upcrossing lemma and Doob's convergence theorem.
- Durrett [2019], Sections 5.2–5.4: martingale convergence in the context of probability theory, with the full upcrossing proof.
- Doob [1953], Chapter VII: the original treatment of martingale convergence, including the upcrossing inequality.
- Robbins and Siegmund [1971]: the original paper proving the almost-supermartingale convergence theorem used throughout stochastic approximation.
- Shiryaev [2016], Chapter VII, Sections 3–4: submartingale convergence and its applications.
- Grimmett and Stirzaker [2001], Chapter 12: martingale convergence with applications to branching processes and random walks.

3 Submartingales and Supermartingales

We recall the definitions from Lesson 8 (Definition 5.4) and develop their key properties.

Definition 3.1 (Submartingale and Supermartingale)**.** Let $\{M_n\}_{n\geq 0}$ be adapted to a filtration $\{\mathcal{F}_n\}$ with $\mathbb{E}[|M_n|] < \infty$ for all n.

- $\{M_n\}$ is a *submartingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ a.s. for all n.
- $\{M_n\}$ is a *supermartingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \leq M_n$ a.s. for all n.

A martingale is both a submartingale and a supermartingale.

Remark 3.2 (Monotonicity of expectations). If $\{M_n\}$ is a submartingale, then $\mathbb{E}[M_n]$ is non-decreasing in n: by the tower property, $\mathbb{E}[M_{n+1}] = \mathbb{E}[\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]] \geq \mathbb{E}[M_n]$. Symmetrically, if $\{M_n\}$ is a supermartingale, then $\mathbb{E}[M_n]$ is non-increasing. A supermartingale models a "losing game" (expected value decreases), while a submartingale models a "winning game."

Remark 3.3 (Duality). $\{M_n\}$ is a supermartingale if and only if $\{-M_n\}$ is a submartingale. This duality allows us to state and prove results for submartingales and immediately obtain the corresponding supermartingale versions.

4 Convex Functions of Martingales

Convex Functions of Martingales

Proposition 4.1 (Convex Functions of Martingales)**.** *Let $\{M_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$, and let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function with $\mathbb{E}[|\varphi(M_n)|] < \infty$ for all n. Then $\{\varphi(M_n)\}$ is a submartingale.*

Proof. We verify the three conditions. **(i)** Since M_n is $\mathcal{F}_n$-measurable and φ is a Borel-measurable function, $\varphi(M_n)$ is $\mathcal{F}_n$-measurable. **(ii)** By assumption, $\mathbb{E}[|\varphi(M_n)|] < \infty$. **(iii)** By the conditional Jensen's inequality (Lesson 9, Theorem 5.1):

$$\mathbb{E}[\varphi(M_{n+1}) \mid \mathcal{F}_n] \geq \varphi(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]) = \varphi(M_n), \tag{1}$$

where the equality uses the martingale property $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$. □

Corollary 4.2 (Squared Martingale is a Submartingale)**.** *If $\{M_n\}$ is a martingale with $\mathbb{E}[M_n^2] < \infty$ for all n, then $\{M_n^2\}$ is a submartingale.*

Proof. Apply Proposition 4.1 with $\varphi(x) = x^2$, which is convex on $\mathbb{R}$. The integrability condition $\mathbb{E}[|M_n^2|] = \mathbb{E}[M_n^2] < \infty$ holds by assumption. □

Remark 4.3. Since $\{M_n^2\}$ is a submartingale, $\mathbb{E}[M_n^2] \geq \mathbb{E}[M_0^2]$. If $M_0 = 0$, then $\mathrm{Var}(M_n) = \mathbb{E}[M_n^2]$ is non-decreasing. This quantifies the intuition that the "spread" of a martingale can only grow over time. In RL, this means that the variance of accumulated TD errors (which form a martingale) grows with the number of updates—motivating the use of decreasing step sizes to counteract this growth.

Example 4.4 (Absolute value and exponential)**.** Further consequences of Proposition 4.1:

1. $\varphi(x) = |x|$ is convex, so if $\{M_n\}$ is a martingale with $\mathbb{E}[|M_n|] < \infty$, then $\{|M_n|\}$ is a submartingale.
2. For any $\lambda \in \mathbb{R}$, $\varphi(x) = e^{\lambda x}$ is convex. If $\mathbb{E}[e^{\lambda M_n}] < \infty$, then $\{e^{\lambda M_n}\}$ is a submartingale. This connects to the exponential supermartingale used in the proof of the Azuma–Hoeffding inequality (Lesson 9, Theorem 4.1).

5 Doob's Upcrossing Inequality

The upcrossing inequality is the key technical tool for proving martingale convergence. It bounds the expected number of times a process can oscillate upward across an interval.

Definition 5.1 (Upcrossings)**.** For a sequence of real numbers $x_0, x_1, \ldots, x_N$ and real numbers $a < b$, an *upcrossing* of the interval $[a, b]$ is a pair of times $\sigma < \tau$ such that $x_\sigma \leq a$ and $x_\tau \geq b$, with no intermediate upcrossing completed between σ and τ.

Formally, define stopping times inductively: $\sigma_1 = \min\{n : x_n \leq a\}$, $\tau_1 = \min\{n > \sigma_1 : x_n \geq b\}$, and for $k \geq 2$, $\sigma_k = \min\{n > \tau_{k-1} : x_n \leq a\}$, $\tau_k = \min\{n > \sigma_k : x_n \geq b\}$ (with $\min \emptyset = \infty$). The *number of upcrossings* of $[a, b]$ by $x_0, \ldots, x_N$ is

$$U_N([a, b]) = \max\{k \geq 0 : \tau_k \leq N\}. \tag{2}$$

Remark 5.2. Each upcrossing represents the process crossing from at or below a to at or above b. If the sequence oscillates between the two levels, the number of upcrossings counts how many complete upward crossings occur. A sequence that converges cannot oscillate indefinitely, so bounding the expected number of upcrossings bounds the failure to converge.

Doob's Upcrossing Inequality

Theorem 5.3 (Doob's Upcrossing Inequality)**.** *Let $\{M_n\}_{n=0}^N$ be a submartingale and let $a < b$. Then*

$$\mathbb{E}[U_N([a, b])] \leq \frac{\mathbb{E}[(M_N - a)^+]}{b - a} \leq \frac{\mathbb{E}[|M_N|] + |a|}{b - a}, \tag{3}$$

where $x^+ = \max(x, 0)$.

Proof. **Step 1: Reduction to non-negative submartingale.** Define $\tilde{M}_n = (M_n - a)^+$. The function $\varphi(x) = (x-a)^+$ is convex and non-decreasing, so if $\{M_n\}$ is a submartingale, then $\{\tilde{M}_n\}$ is also a submartingale (this follows from a version of conditional Jensen: for non-decreasing convex φ and submartingale $\{M_n\}$, $\mathbb{E}[\varphi(M_{n+1}) \mid \mathcal{F}_n] \geq \varphi(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]) \geq \varphi(M_n)$, where the last inequality uses $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ and the monotonicity of φ).

Moreover, $\tilde{M}_n \geq 0$, and the number of upcrossings of $[a, b]$ by $\{M_n\}$ equals the number of upcrossings of $[0, b - a]$ by $\{\tilde{M}_n\}$. So it suffices to prove the inequality for a non-negative submartingale $\{Y_n\}$ upcrossing $[0, c]$ where $c = b - a > 0$.

Step 2: Gambling strategy argument. Define a *betting strategy* H_n ($n \geq 1$) by: bet $H_n = 1$ ("in the game") when trying to complete an upcrossing (i.e., during the interval from σ_k to τ_k for some k), and $H_n = 0$ ("out of the game") otherwise. Formally, define the stopping times σ_k, τ_k as in Definition 5.1 for the sequence $\{Y_n\}$ and the interval $[0, c]$, and set

$$H_n = \sum_{k=1}^{\infty} \mathbf{1}_{\{\sigma_k < n \leq \tau_k\}}. \tag{4}$$

Since σ_k and τ_k are stopping times with respect to $\{\mathcal{F}_n\}$, the event $\{\sigma_k < n \leq \tau_k\} = \{\sigma_k \leq n - 1\} \cap \{\tau_k \geq n\}$ is in $\mathcal{F}_{n-1}$ (as it depends only on $Y_0, \ldots, Y_{n-1}$). Therefore H_n is predictable.

Step 3: Accounting for gains. Define the "gains process"

$$G_N = \sum_{n=1}^{N} H_n(Y_n - Y_{n-1}). \tag{5}$$

Each completed upcrossing contributes a gain of at least c (the process goes from ≤ 0 to $\geq c$), so $G_N \geq c \cdot U_N([0, c])$. Between upcrossings, $H_n = 0$, so those increments do not contribute to G_N. The final incomplete upcrossing (if any) contributes a non-negative amount (since $Y_n \geq 0$,

the process can only go down to 0, contributing $\geq -0 = 0$ to the sum if we started above 0; more precisely, during an incomplete upcrossing the sum of increments $\geq Y_N - Y_{\sigma_k} \geq 0 - 0 = 0$ is not necessarily non-negative, but we account for this below).

We instead write:

$$G_N = \sum_{n=1}^{N} H_n(Y_n - Y_{n-1}) = \sum_{n=1}^{N}(Y_n - Y_{n-1}) - \sum_{n=1}^{N}(1 - H_n)(Y_n - Y_{n-1}). \tag{6}$$

The first sum telescopes to $Y_N - Y_0$. The second sum represents the gains when the strategy is "out": define $G_N^{\text{out}} = \sum_{n=1}^{N}(1 - H_n)(Y_n - Y_{n-1})$.

Step 4: Expectation bound. Since $\{Y_n\}$ is a non-negative submartingale and $(1 - H_n)$ is a non-negative predictable process, the "out" gains form a submartingale transform. A key property of submartingales is that predictable non-negative transforms preserve the submartingale property (this is the discrete-time stochastic integral inequality): for any non-negative predictable $\{C_n\}$,

$$\mathbb{E}\left[\sum_{n=1}^{N} C_n(Y_n - Y_{n-1})\right] \geq 0 \tag{7}$$

when $\{Y_n\}$ is a submartingale. To see this, note that $\mathbb{E}[C_n(Y_n - Y_{n-1}) \mid \mathcal{F}_{n-1}] = C_n\mathbb{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}] \geq 0$ since C_n is $\mathcal{F}_{n-1}$-measurable and $\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] \geq Y_{n-1}$. Taking unconditional expectations and summing: $\mathbb{E}[G_N^{\text{out}}] \geq 0$.

From (6): $G_N = (Y_N - Y_0) - G_N^{\text{out}}$, so $\mathbb{E}[G_N] = \mathbb{E}[Y_N] - \mathbb{E}[Y_0] - \mathbb{E}[G_N^{\text{out}}] \leq \mathbb{E}[Y_N]$ (using $Y_0 \geq 0$ and $\mathbb{E}[G_N^{\text{out}}] \geq 0$).

Since $G_N \geq c \cdot U_N([0, c])$:

$$c \cdot \mathbb{E}[U_N([0, c])] \leq \mathbb{E}[G_N] \leq \mathbb{E}[Y_N] = \mathbb{E}[(M_N - a)^+]. \tag{8}$$

Dividing by $c = b - a$ gives the first inequality in (3).

Step 5: Second inequality. Since $(M_N - a)^+ \leq |M_N - a| \leq |M_N| + |a|$, we have $\mathbb{E}[(M_N - a)^+] \leq \mathbb{E}[|M_N|] + |a|$, giving the second inequality. □

6 Doob's Forward Convergence Theorem

Doob's Forward Convergence Theorem

Theorem 6.1 (Doob's Forward Convergence Theorem). *Let $\{M_n\}_{n\geq 0}$ be a submartingale with $\sup_n \mathbb{E}[M_n^+] < \infty$. Then the limit $M_\infty = \lim_{n\to\infty} M_n$ exists and is finite almost surely.*
In particular, every non-negative supermartingale converges almost surely to a finite limit.

Proof. **Step 1: Upcrossings are finite a.s.** The limit $\lim_{n\to\infty} M_n$ fails to exist as a value in $[-\infty, +\infty]$ if and only if $\liminf_n M_n < \limsup_n M_n$, which occurs if and only if M_n performs infinitely many upcrossings of some rational interval $[a, b]$ with $a < b$:

$$\{\lim M_n \text{ does not exist}\} = \bigcup_{\substack{a,b\in\mathbb{Q} \\ a<b}} \{U_\infty([a, b]) = \infty\}. \tag{9}$$

For each $a, b \in \mathbb{Q}$ with $a < b$, the upcrossing inequality (Theorem 5.3) gives

$$\mathbb{E}[U_N([a, b])] \leq \frac{\mathbb{E}[M_N^+] + |a|}{b - a} \leq \frac{\sup_n \mathbb{E}[M_n^+] + |a|}{b - a} < \infty. \tag{10}$$

Since $U_N([a,b])$ is non-decreasing in N, the monotone convergence theorem (Lesson 11, Theorem 3.1) gives

$$\mathbb{E}[U_\infty([a,b])] = \lim_{N\to\infty} \mathbb{E}[U_N([a,b])] \le \frac{\sup_n \mathbb{E}[M_n^+] + |a|}{b-a} < \infty. \tag{11}$$

Since $U_\infty([a,b])$ has finite expectation, it is finite almost surely: $\mathbb{P}(U_\infty([a,b]) = \infty) = 0$.

Step 2: The limit exists a.s. By (9) and a countable union over $a, b \in \mathbb{Q}$:

$$\mathbb{P}(\lim M_n \text{ does not exist}) \le \sum_{\substack{a,b\in\mathbb{Q}\\ a<b}} \mathbb{P}(U_\infty([a,b]) = \infty) = 0. \tag{12}$$

So $M_\infty = \lim_{n\to\infty} M_n$ exists a.s. (as a value in $[-\infty, +\infty]$).

Step 3: The limit is finite a.s. We must show $|M_\infty| < \infty$ a.s. Since $\{M_n\}$ is a submartingale, $\mathbb{E}[M_n]$ is non-decreasing (Remark 3.2), so $\mathbb{E}[M_n] \ge \mathbb{E}[M_0] > -\infty$ for all n. Now $\mathbb{E}[M_n^+] \le C := \sup_n \mathbb{E}[M_n^+] < \infty$ by hypothesis. By Fatou's lemma (Lesson 11, Theorem 3.4):

$$\mathbb{E}[M_\infty^+] \le \liminf_{n\to\infty} \mathbb{E}[M_n^+] \le C < \infty. \tag{13}$$

For the negative part, using $|M_n| = M_n^+ + M_n^-$ and $\mathbb{E}[M_n] = \mathbb{E}[M_n^+] - \mathbb{E}[M_n^-]$, we get $\mathbb{E}[M_n^-] = \mathbb{E}[M_n^+] - \mathbb{E}[M_n] \le C - \mathbb{E}[M_0]$. By Fatou's lemma applied to $M_n^- \ge 0$:

$$\mathbb{E}[M_\infty^-] \le \liminf_{n\to\infty} \mathbb{E}[M_n^-] \le C - \mathbb{E}[M_0] < \infty. \tag{14}$$

Therefore $\mathbb{E}[|M_\infty|] = \mathbb{E}[M_\infty^+] + \mathbb{E}[M_\infty^-] < \infty$, which implies $|M_\infty| < \infty$ a.s.

Step 4: Non-negative supermartingale case. Let $\{M_n\}$ be a non-negative supermartingale. Then $M_n^+ = M_n$ (since $M_n \ge 0$), and $\mathbb{E}[M_n^+] = \mathbb{E}[M_n] \le \mathbb{E}[M_0] < \infty$ (since expectations are non-increasing for supermartingales). Also $\{-M_n\}$ is a submartingale (Remark 3.3) with $\sup_n \mathbb{E}[(-M_n)^+] = \sup_n \mathbb{E}[M_n^-] = 0$ (since $M_n \ge 0$). But we can apply the theorem directly: $\{M_n\}$ being a non-negative supermartingale means $M_n \ge 0$ and $M_n^+ = M_n$, so $\sup_n \mathbb{E}[M_n^+] = \sup_n \mathbb{E}[M_n] \le \mathbb{E}[M_0] < \infty$. We need $\{M_n\}$ to be a submartingale for the theorem, so instead note that $\{-M_n\}$ is a submartingale with $\sup_n \mathbb{E}[(-M_n)^+] = \sup_n \mathbb{E}[\max(-M_n, 0)] = 0 < \infty$ (since $M_n \ge 0$ implies $-M_n \le 0$). By the theorem, $-M_n$ converges a.s., hence M_n converges a.s. to a finite limit. □

Non-Negative Supermartingales in RL

The non-negative supermartingale convergence result is the workhorse in RL convergence proofs. The standard approach is:

1. Define a Lyapunov function $L_n \ge 0$ measuring distance from the optimum (e.g., $L_n = \|V_n - V^\pi\|^2$ or $L_n = \|Q_n - Q^*\|_\infty$).
2. Show $\mathbb{E}[L_{n+1} \mid \mathcal{F}_n] \le L_n$ (or approximately so, with summable perturbations handled by the Robbins–Siegmund theorem).
3. Conclude $L_n \to L_\infty$ a.s. by Doob's theorem.
4. Use additional drift analysis to show $L_\infty = 0$.

Example 6.2 (Bounded martingale convergence)**.** Let $\{M_n\}$ be a martingale with $|M_n| \le C$ a.s. for all n and some constant C. Then $M_n^+ \le C$, so $\sup_n \mathbb{E}[M_n^+] \le C < \infty$. Since every martingale is a submartingale, Doob's theorem applies: $M_n \to M_\infty$ a.s. with $|M_\infty| \le C$ a.s.

More generally, any L^1-bounded martingale (i.e., $\sup_n \mathbb{E}[|M_n|] < \infty$) converges a.s., since $\mathbb{E}[M_n^+] \le \mathbb{E}[|M_n|] \le \sup_n \mathbb{E}[|M_n|] < \infty$.

7 The Robbins–Siegmund Theorem

Doob's convergence theorem handles exact supermartingales. In practice, the processes arising in stochastic approximation are only *approximately* supermartingales: the supermartingale inequality holds up to summable perturbation terms. The Robbins–Siegmund theorem provides the precise generalization needed.

Robbins–Siegmund Theorem

Theorem 7.1 (Robbins–Siegmund (1971)). *Let $\{\mathcal{F}_n\}_{n\geq 0}$ be a filtration and let $\{V_n\}$, $\{\alpha_n\}$, $\{\beta_n\}$, $\{\gamma_n\}$ be sequences of non-negative random variables adapted to $\{\mathcal{F}_n\}$ satisfying*

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1+\alpha_n)V_n - \beta_n + \gamma_n \quad \text{a.s. for all } n \geq 0, \tag{15}$$

with $\sum_{n=0}^{\infty} \alpha_n < \infty$ and $\sum_{n=0}^{\infty} \gamma_n < \infty$ almost surely. Then:

1. *V_n converges a.s. to a finite non-negative random variable $V_\infty \geq 0$.*
2. *$\sum_{n=0}^{\infty} \beta_n < \infty$ almost surely.*

Proof. **Step 1: Construct a non-negative supermartingale.** Since $\sum_{n=0}^{\infty} \gamma_n < \infty$ a.s., define the tail sum $\Gamma_n = \sum_{k=n}^{\infty} \gamma_k$, which is finite a.s., non-negative, and non-increasing: $\Gamma_{n+1} = \Gamma_n - \gamma_n$. Define

$$U_n = V_n + \Gamma_n. \tag{16}$$

Then $U_n \geq V_n \geq 0$ and

$$\begin{aligned}
\mathbb{E}[U_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[V_{n+1} \mid \mathcal{F}_n] + \Gamma_{n+1} \\
&\leq (1+\alpha_n)V_n - \beta_n + \gamma_n + \Gamma_{n+1} \\
&= (1+\alpha_n)V_n - \beta_n + \gamma_n + (\Gamma_n - \gamma_n) \\
&= (1+\alpha_n)V_n - \beta_n + \Gamma_n \\
&\leq (1+\alpha_n)V_n + \Gamma_n \\
&\leq (1+\alpha_n)(V_n + \Gamma_n) \\
&= (1+\alpha_n)U_n.
\end{aligned} \tag{17}$$

The penultimate inequality uses $\alpha_n \geq 0$ and $\Gamma_n \geq 0$: $(1+\alpha_n)V_n + \Gamma_n \leq (1+\alpha_n)V_n + (1+\alpha_n)\Gamma_n$.

Step 2: Normalize to an exact supermartingale. Define $A_n = \prod_{k=0}^{n-1}(1+\alpha_k)$ for $n \geq 1$ and $A_0 = 1$. Since $\sum_k \alpha_k < \infty$ a.s., the infinite product converges:

$$A_\infty = \prod_{k=0}^{\infty}(1+\alpha_k) \leq \exp\left(\sum_{k=0}^{\infty} \alpha_k\right) < \infty \quad \text{a.s.} \tag{18}$$

(using the inequality $1 + x \leq e^x$ for $x \geq 0$). Define $\tilde{U}_n = U_n/A_n$. Since $A_n > 0$ and $U_n \geq 0$, we have $\tilde{U}_n \geq 0$. From (17):

$$\mathbb{E}[\tilde{U}_{n+1} \mid \mathcal{F}_n] = \frac{\mathbb{E}[U_{n+1} \mid \mathcal{F}_n]}{A_{n+1}} \leq \frac{(1+\alpha_n)U_n}{A_{n+1}} = \frac{U_n}{A_n} = \tilde{U}_n, \tag{19}$$

since $A_{n+1} = (1+\alpha_n)A_n$. So $\{\tilde{U}_n\}$ is a non-negative supermartingale.

Step 3: Apply Doob's convergence theorem. By Theorem 6.1 (non-negative supermartingale case), $\tilde{U}_n \to \tilde{U}_\infty$ a.s. with $\tilde{U}_\infty$ finite.

Since $A_n \to A_\infty$ with $0 < A_\infty < \infty$ a.s.:

$$U_n = A_n\tilde{U}_n \to A_\infty\tilde{U}_\infty \quad \text{a.s.}, \tag{20}$$

and the limit is finite. Since $\Gamma_n \to 0$ a.s. (as the tail of a convergent series):

$$V_n = U_n - \Gamma_n \to A_\infty \tilde{U}_\infty - 0 = V_\infty \quad \text{a.s.}, \tag{21}$$

proving conclusion (1).

Step 4: Summability of β_n. From (17), keeping the β_n term:

$$\mathbb{E}[U_{n+1} \mid \mathcal{F}_n] \le (1+\alpha_n)V_n - \beta_n + \Gamma_n \le (1+\alpha_n)U_n - \beta_n. \tag{22}$$

Rearranging: $\beta_n \le (1+\alpha_n)U_n - \mathbb{E}[U_{n+1} \mid \mathcal{F}_n]$. Summing from $n=0$ to N and taking expectations:

$$\begin{aligned}
\mathbb{E}\left[\sum_{n=0}^{N} \beta_n\right] &\le \sum_{n=0}^{N}(1+\alpha_n)\mathbb{E}[U_n] - \sum_{n=0}^{N}\mathbb{E}[U_{n+1}] \\
&= \sum_{n=0}^{N}\alpha_n\mathbb{E}[U_n] + \mathbb{E}[U_0] - \mathbb{E}[U_{N+1}] \\
&\le \sum_{n=0}^{N}\alpha_n\mathbb{E}[U_n] + \mathbb{E}[U_0].
\end{aligned} \tag{23}$$

Since $\tilde{U}_n$ is a non-negative supermartingale, $\mathbb{E}[\tilde{U}_n] \le \mathbb{E}[\tilde{U}_0] = \mathbb{E}[U_0]$, so $\mathbb{E}[U_n] = \mathbb{E}[A_n\tilde{U}_n] \le A_\infty\mathbb{E}[U_0]$ (on the event where $A_\infty < \infty$, which has probability 1). Therefore $\sum_{n=0}^{N}\alpha_n\mathbb{E}[U_n] \le A_\infty\mathbb{E}[U_0]\sum_{n=0}^{\infty}\alpha_n < \infty$. So $\mathbb{E}[\sum_{n=0}^{\infty}\beta_n] < \infty$, which implies $\sum_{n=0}^{\infty}\beta_n < \infty$ a.s., proving conclusion (2). □

Robbins–Siegmund in TD Convergence

The Robbins–Siegmund theorem is the primary convergence tool for stochastic approximation in RL. In the TD(0) convergence proof, the identification is:

- $V_n = \|V_n - V^\pi\|^2$: the Lyapunov function (squared error between the current estimate and the true value function).
- $\alpha_n = O(\alpha_n^2)$: arises from the squared step-size term after expanding the update.
- $\beta_n = c\,\alpha_n\|V_n - V^\pi\|^2$: the drift toward the fixed point, arising from the contraction property of the Bellman operator.
- $\gamma_n = C\,\alpha_n^2$: arises from the noise variance (the MDS squared term bounded using Lesson 8, Theorem 7.1).

The step-size conditions $\sum\alpha_n = \infty$ and $\sum\alpha_n^2 < \infty$ (Robbins–Monro conditions) ensure $\sum\gamma_n < \infty$ and $\sum\alpha_n < \infty$ (for the rescaled version). Conclusion (2) gives $\sum_n \alpha_n\|V_n - V^\pi\|^2 < \infty$, and since $\sum\alpha_n = \infty$, this forces $\liminf_n \|V_n - V^\pi\| = 0$. Combined with conclusion (1), this gives $V_n \to V^\pi$ a.s.

Remark 7.2 (Role of each condition). The conditions in the Robbins–Siegmund theorem have clear interpretations:

- $\sum\alpha_n < \infty$: the multiplicative perturbation $(1+\alpha_n)$ is "nearly 1" in a summable sense.
- $\sum\gamma_n < \infty$: the additive noise is summable.
- $\beta_n \ge 0$: there is a non-negative "drift" term that the process is sacrificing. Conclusion (2) says this drift is summable, meaning the process eventually stops drifting.

8 Exercises

Exercise 8.1 ((*)). Let $X_n = 1/(n+1)$ for $n \ge 0$. Verify that $\{X_n\}$ is a non-negative supermartingale with respect to the trivial filtration $\mathcal{F}_n = \{\emptyset, \Omega\}$ and that it converges a.s., consistent with Doob's theorem.

Exercise 8.2 ((*)). Let $\{M_n\}$ be a martingale with $\mathbb{E}[e^{\lambda|M_n|}] < \infty$ for some $\lambda > 0$. Show that $\{e^{\lambda M_n}\}$, $\{e^{-\lambda M_n}\}$, and $\{\cosh(\lambda M_n)\}$ are all submartingales.

Exercise 8.3 ((**)). Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}[X_i] = 1$, $X_i \geq 0$, and $\mathbb{P}(X_i = 0) > 0$. Define $M_n = \prod_{k=1}^{n} X_k$. Show that $\{M_n\}$ is a non-negative martingale. By Doob's theorem, $M_n \to M_\infty$ a.s. Show that $M_\infty = 0$ a.s. *Hint:* Use the second Borel–Cantelli lemma (Lesson 10, Theorem 5.2) to show $X_k = 0$ infinitely often.

Exercise 8.4 ((**)). Let $X_0, X_1, \ldots, X_8$ be the sequence $1, 3, 0, 4, 1, 5, 2, 0, 3$. Compute $U_8([1, 4])$ by identifying all upcrossings of the interval $[1, 4]$. Verify that the count is consistent with the upcrossing inequality applied with $a = 1$, $b = 4$.

Exercise 8.5 ((**)). Let $V_n = V_0 \cdot \prod_{k=0}^{n-1}(1 - c\alpha_k + w_k)$ where $c > 0$, $\alpha_k \geq 0$, $\sum \alpha_k = \infty$, $\sum \alpha_k^2 < \infty$, and $\{w_k\}$ are i.i.d. with $\mathbb{E}[w_k] = 0$, $|w_k| \leq D\alpha_k$. Show that $\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1 + C\alpha_n^2)V_n - c\alpha_n V_n + C'\alpha_n^2$ for appropriate constants. Identify the roles of α_n, β_n, and γ_n in the Robbins–Siegmund framework and conclude $V_n \to 0$ a.s.

Exercise 8.6 ((***)). Construct a non-negative martingale $\{M_n\}$ with $M_0 = 1$ that converges a.s. to $M_\infty = 0$. Conclude that $M_n \xrightarrow{\text{a.s.}} M_\infty$ but $\mathbb{E}[M_\infty] = 0 \neq 1 = \mathbb{E}[M_n]$, showing that Doob's convergence theorem gives a.s. convergence but not L^1 convergence in general. *Hint:* Consider $M_n = 2^n \mathbf{1}_{A_n}$ where $\{A_n\}$ is a decreasing sequence of events with $\mathbb{P}(A_n) = 2^{-n}$.

Exercise 8.7 ((***)). **(Two-timescale SA preview.)** Consider two sequences satisfying $\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1 + a_n)V_n + b_n W_n + c_n$ and $\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] \leq (1 + d_n)W_n + e_n$, where $\sum a_n, \sum c_n, \sum d_n, \sum e_n < \infty$ and $b_n \geq 0$ with $\sum b_n < \infty$. Apply the Robbins–Siegmund theorem first to $\{W_n\}$ (showing W_n converges and W_n is bounded a.s.), then to $\{V_n\}$. Conclude that both sequences converge a.s. This models the two-timescale SA structure in actor-critic algorithms.

9 Summary

- A **submartingale** satisfies $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ (non-decreasing expectations); a **supermartingale** satisfies the reverse inequality.
- **Convex functions of martingales are submartingales** (Proposition 4.1). In particular, $\{M_n^2\}$ and $\{|M_n|\}$ are submartingales when $\{M_n\}$ is a martingale.
- **Doob's upcrossing inequality** (Theorem 5.3): the expected number of upcrossings of $[a, b]$ by a submartingale up to time N is at most $(\mathbb{E}[M_N^+] + |a|)/(b - a)$.
- **Doob's forward convergence theorem** (Theorem 6.1): a submartingale with $\sup_n \mathbb{E}[M_n^+] < \infty$ converges a.s. to a finite limit. Every non-negative supermartingale converges a.s.
- **Robbins–Siegmund theorem** (Theorem 7.1): if $\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1 + \alpha_n)V_n - \beta_n + \gamma_n$ with $\sum \alpha_n, \sum \gamma_n < \infty$, then V_n converges a.s. and $\sum \beta_n < \infty$ a.s.
- The Robbins–Siegmund theorem is the **master convergence tool** for stochastic approximation in RL, handling the perturbations that prevent Lyapunov functions from being exact supermartingales.

10 Further Reading

- **Lesson 8** defines martingales, MDS, and submartingales/supermartingales—the objects whose convergence is analyzed in this lesson.
- **Lesson 9** provides the probability inequalities (Jensen, Markov) used in the proofs, and the Azuma–Hoeffding inequality which gives *rate* information complementing the a.s. convergence established here.

- **Lesson 14** applies the Robbins–Siegmund theorem to prove the TD noise decomposition, showing that TD updates fit the stochastic approximation framework.
- Williams [1991], Chapters 11–12: the most accessible presentation of Doob's convergence theorem.
- Durrett [2019], Section 5.2: the upcrossing proof of martingale convergence with additional applications.
- Robbins and Siegmund [1971]: the original Robbins–Siegmund paper, concise and readable.
- Doob [1953]: the foundational text on martingale theory and convergence.

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